

Counting(计数)

Chapter 6

The Basics of Counting

Section 6.1

6.1.2 Basic Counting Principles: The Product Rule

The Product Rule: A procedure can be broken down into a sequence of two tasks. There are n_1 ways to do the first task and n_2 ways to do the second task. Then there are $n_1 \cdot n_2$ ways to do the procedure.

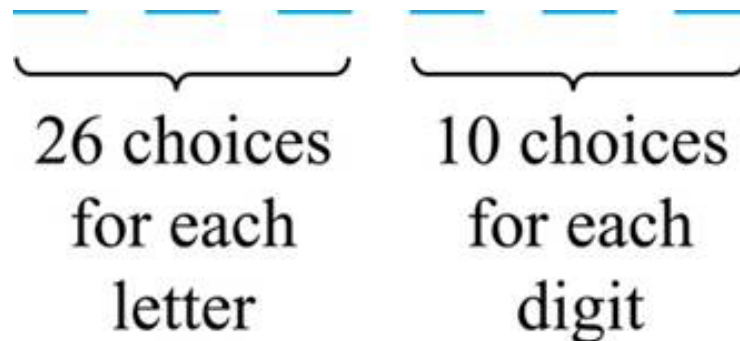
Example 4: How many bit strings(位串) of length seven are there?

Solution: Since each of the seven bits is either a 0 or a 1, the answer is $2^7 = 128$.

The Product Rule

Example 5: How many different license plates(车牌) can be made if each plate contains a sequence of three uppercase English letters followed by three digits?

Solution: By the product rule,
there are $26 \cdot 26 \cdot 26 \cdot 10 \cdot 10 \cdot 10 = 17,576,000$
different possible license plates.



[Jump to long description](#)

Counting Subsets of a Finite Set

Example 10: Counting Subsets of a Finite Set: Use the product rule to show that the number of different subsets of a finite set S is $2^{|S|}$. (*In Section 5.1, mathematical induction was used to prove this same result.*)

Solution: When the elements of S are listed in an arbitrary order, there is a one-to-one correspondence between **subsets of S** and **bit strings(位串) of length $|S|$** . When the i th element is in the subset, the bit string has a 1 in the i th position and a 0 otherwise.

By the product rule, there are $2^{|S|}$ such bit strings, and therefore $2^{|S|}$ subsets.

Definition: The **cardinality** (基数) of a finite set A , denoted by $|A|$, is the number of (distinct) elements of A .

Basic Counting Principles:

The Sum Rule

The Sum Rule: If a task can be done either in one of n_1 ways or in one of n_2 , where none of the set of n_1 ways is the same as any of the n_2 ways, then there are $n_1 + n_2$ ways to do the task.

Example 12: The mathematics department must choose either a student or a faculty member as a representative for a university committee. How many choices are there for this representative if there are 37 members of the mathematics faculty and 83 mathematics majors and no one is both a faculty member and a student.

Solution: By the sum rule it follows that there are $37 + 83 = 120$ possible ways to pick a representative.

The Sum Rule in terms of sets.

The sum rule can be phrased in terms of sets.

$|A \cup B| = |A| + |B|$ as long as A and B are disjoint sets.

Or more generally,

$$|A_1 \cup A_2 \cup \cdots \cup A_m| = |A_1| + |A_2| + \cdots + |A_m|$$

when $A_i \cap A_j = \emptyset$ for all i, j .

Combining the Sum and Product Rule

Example: Suppose statement labels in a programming language can be either a single uppercase letter or a uppercase letter followed by a digit. Find the number of possible labels.

Solution: Use the product rule.

$$26 + 26 \cdot 10 = 286$$

6.1.3 More Complex Counting Problems

Combining the sum and product rule allows us to solve more complex problems.

Example 16 Counting Passwords: Each user on a computer system has a password, which is six to eight characters long, where each character is an uppercase letter or a digit. Each password must contain at least one digit. How many possible passwords are there?

Solution: Let P be the total number of passwords, and let P_6 , P_7 , and P_8 be the passwords of length 6, 7, and 8.

- By the sum rule $P = P_6 + P_7 + P_8$.
- To find each of P_6 , P_7 , and P_8 , we find the number of passwords of the specified length composed of letters and digits and subtract the number composed only of letters. We find that:

$$P_6 = 36^6 - 26^6 = 2,176,782,336 - 308,915,776 = 1,867,866,560.$$

$$P_7 = 36^7 - 26^7 =$$

$$78,364,164,096 - 8,031,810,176 = 70,332,353,920.$$

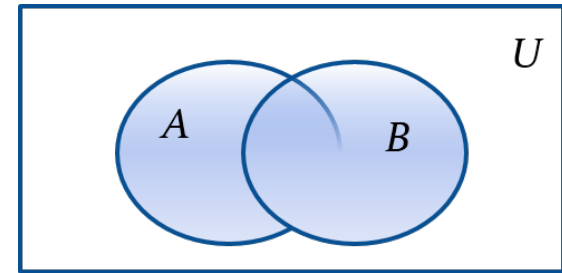
$$P_8 = 36^8 - 26^8 =$$

$$2,821,109,907,456 - 208,827,064,576 = 2,612,282,842,880.$$

Consequently, $P = P_6 + P_7 + P_8 = 2,684,483,063,360$.

6.1.4 Basic Counting Principles: Subtraction Rule

Subtraction Rule(減法法則): If a task can be done **either** in one of n_1 ways **or** in one of n_2 ways, then the total number of ways to do the task is $n_1 + n_2$ **minus** the number of ways to do the task that are **common** to the two different ways.



Also known as, the *principle of inclusion-exclusion*(包含排除):

$$|A \cup B| = |A| + |B| - |A \cap B|$$

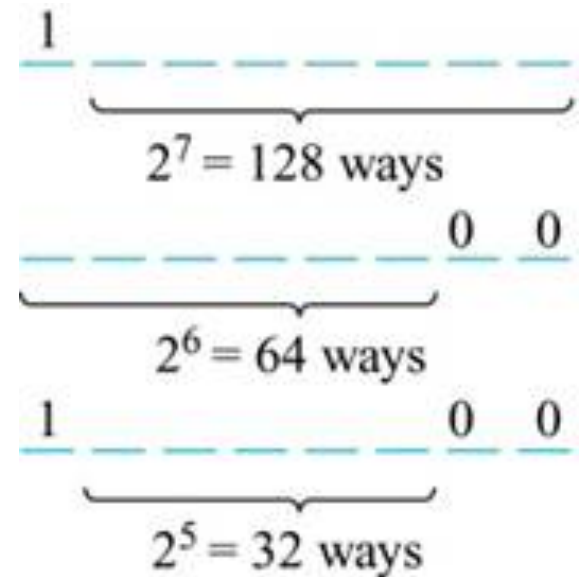
This is the formula given in Section 2.2 for the number of elements in the union of two sets.

Counting Bit Strings

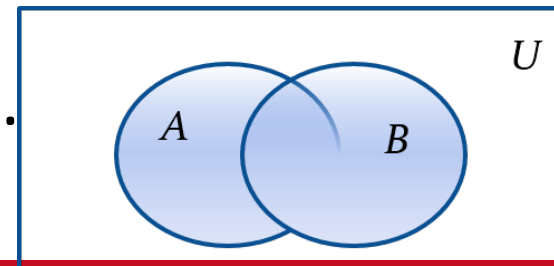
Example 18: How many bit strings of length eight **either** start with a 1 bit **or** end with the two bits 00?

Solution: Use the subtraction rule.

- Number of bit strings of length eight that start with a 1 bit: $2^7 = 128$
- Number of bit strings of length eight that end with bits 00: $2^6 = 64$
- Number of bit strings of length eight that start with a 1 bit and end with bits 00 : $2^5 = 32$



Hence, the number is $128 + 64 - 32 = 160$.



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Section Summary₁

The Product Rule: A procedure can be broken down into a sequence of two tasks. There are n_1 ways to do the first task and n_2 ways to do the second task. Then there are $n_1 \cdot n_2$ ways to do the procedure.

The Sum Rule: If a task can be done either in one of n_1 ways or in one of n_2 , where none of the set of n_1 ways is the same as any of the n_2 ways, then there are $n_1 + n_2$ ways to do the task.

The Subtraction Rule:

$$|A \cup B| = |A| + |B| - |A \cap B|$$

10. How many bit strings are there of length eight?

51. How many bit strings of length 10 either begin with three 0s or end with two 0s?

$2^8 = 256$ bit strings.

51. 352

26. How many strings of four decimal digits

a) do not contain the same digit twice?

b) end with an even digit?

c) have exactly three digits that are 9s?

26. a) There are 10 ways to choose the first digit, 9 ways to choose the second, and so on; therefore the answer is $10 \cdot 9 \cdot 8 \cdot 7 = 5040$.

b) There are 10 ways to choose each of the first three digits and 5 ways to choose the last; therefore the answer is $10^3 \cdot 5 = 5000$.

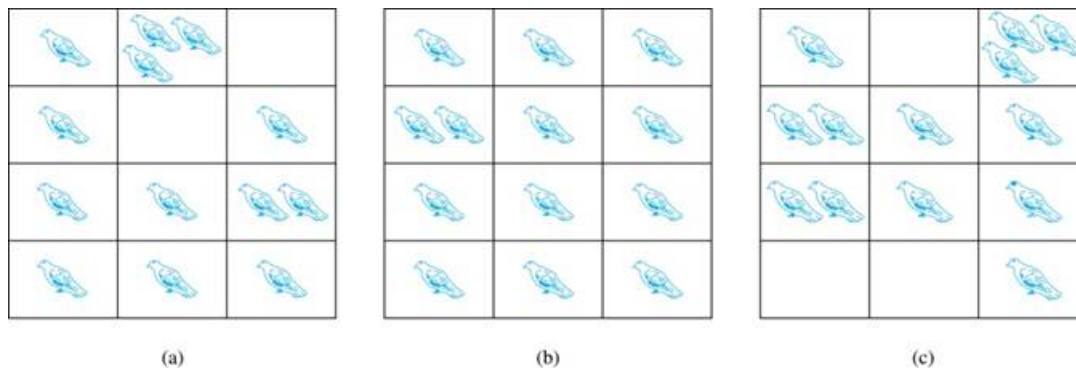
c) There are 4 ways to choose the position that is to be different from 9, and 9 ways to choose the digit to go there. Therefore there are $4 \cdot 9 = 36$ such strings.

The Pigeonhole Principle

Section 6.2

The Pigeonhole(鸽洞) Principle₁

If a flock of 20 pigeons roosts(栖息) in a set of 19 pigeonholes, one of the pigeonholes must have more than 1 pigeon.



Pigeonhole Principle: If k is a positive integer and $k + 1$ or more objects are placed into k boxes, then at least one box contains two or more objects.

Proof: We use a proof by contraposition. Suppose none of the k boxes has more than one object. Then the total number of objects would be at most k . This contradicts the statement that we have $k + 1$ objects.

[Jump to long description](#)

The Pigeonhole Principle₂

Corollary 1: A function f from a set with $k + 1$ elements to a set with k elements is not one-to-one.

Example: Among any group of 367 people, there must be at least two with the same birthday, because there are only 366 possible birthdays.

6.2.2 The Generalized Pigeonhole Principle₁

The Generalized Pigeonhole Principle: If N objects are placed into k boxes, then there is at least one box containing at least $\lceil N/k \rceil$ objects.

Proof: We use a proof by contraposition. Suppose that none of the boxes contains more than $\lceil N/k \rceil - 1$ objects. Then the total number of objects is at most

$$k \left(\left\lceil \frac{N}{k} \right\rceil - 1 \right) < k \left(\left(\frac{N}{k} + 1 \right) - 1 \right) = N,$$

where the inequality $\lceil N/k \rceil < N/k + 1$ has been used. This is a contradiction because there are a total of N objects.

Example: Among 100 people there are at least $\lceil 100/12 \rceil = 9$ who were born in the same month.

The Generalized Pigeonhole Principle₂

- Example:** a) How many cards must be selected from a standard deck of 52 cards to guarantee that at least three cards of the same suit(花色) are chosen?
- b) How many must be selected to guarantee that at least three hearts(红心) are selected?

Solution: a) We assume four boxes; one for each suit. Using the generalized pigeonhole principle, at least one box contains at least $\lceil N/4 \rceil$ cards. At least three cards of one suit are selected if $\lceil N/4 \rceil \geq 3$. The smallest integer N such that $\lceil N/4 \rceil \geq 3$ is $N = 2 \cdot 4 + 1 = 9$.

b) A deck contains 13 hearts and 39 cards which are not hearts. So, if we select 41 cards, we may have 39 cards which are not hearts along with 2 hearts. However, when we select 42 cards, we must have at least three hearts. (Note that the generalized pigeonhole principle is not used here.)

Exercise

2. Show that if there are 30 students in a class, then at least two have last names that begin with the same letter.
3. A drawer contains a dozen brown socks and a dozen black socks, all unmatched. A man takes socks out at random in the dark.
 - a) How many socks must he take out to be sure that he has at least two socks of the same color?
 - b) How many socks must he take out to be sure that he has at least two black socks?

2. This follows from the pigeonhole principle, with $k = 26$.

3 14

Section Summary₂

The Pigeonhole Principle:

If k is a positive integer and $k + 1$ or more objects are placed into k boxes, then at least one box contains two or more objects.

The Generalized Pigeonhole Principle:

If N objects are placed into k boxes, then there is at least one box containing at least $\lceil N/k \rceil$ objects.

Permutations and Combinations

Section 6.3

Permutations

Definition: A *permutation*(排列) of a set of distinct objects is an *ordered* arrangement of these objects. An *ordered* arrangement of r elements of a set is called an *r -permutation*.

Example: Let $S = \{1,2,3\}$.

- The ordered arrangement 3,1,2 is a permutation of S .
- The ordered arrangement 3,2 is a 2-permutation of S .

The *number* of r -permutations of a set with n elements is denoted by $P(n,r)$.

- The 2-permutations of $S = \{1,2,3\}$ are 1,2; 1,3; 2,1; 2,3; 3,1; and 3,2. Hence, $P(3,2) = 6$.

A Formula for the Number of Permutations

Theorem 1: If n is a positive integer and r is an integer with $1 \leq r \leq n$, then there are

$$P(n, r) = n(n - 1)(n - 2) \cdots (n - r + 1)$$

r -permutations of a set with n distinct elements.

Proof: Use the product rule. The first element can be chosen in n ways. The second in $n - 1$ ways, and so on until there are $(n - (r - 1))$ ways to choose the last element.

Note that $P(n, 0) = 1$, since there is only one way to order zero elements.

Corollary 1: If n and r are integers with $1 \leq r \leq n$, then

$$P(n, r) = \frac{n!}{(n - r)!}$$

Solving Counting Problems by Counting Permutations₂

Example 6: Suppose that a saleswoman has to visit eight different cities. She must begin her trip in a specified city, but she can visit the other seven cities in any order she wishes. How many possible orders can the saleswoman use when visiting these cities?

Solution: The first city is chosen, and the rest are ordered arbitrarily. Hence the orders are:

$$7! = 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 5040$$

If she wants to find the tour with the shortest path that visits all the cities, she must consider 5040 paths!

Solving Counting Problems by Counting Permutations₃

Example 7: How many permutations of the letters *ABCDEFGH* contain the string *ABC* ?

Solution: We solve this problem by counting the permutations of six objects, *ABC*, *D*, *E*, *F*, *G*, and *H*.

$$6! = 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 720$$

6.3.3 Combinations₁

Definition: An *r-combination*(组合) of elements of a set is an *unordered* selection of r elements from the set. Thus, an r -combination is simply a subset of the set with r elements. The number of r -combinations of a set with n distinct elements is denoted by $C(n, r)$. The notation $\binom{n}{r}$ is also used and is called a *binomial coefficient*(二项式系数).

Example: Let S be the set $\{a, b, c, d\}$. Then $\{a, c, d\}$ is a 3-combination from S . It is the same as $\{d, c, a\}$ since the order listed does not matter.

$C(4,2) = 6$ because the 2-combinations of $\{a, b, c, d\}$ are the six subsets $\{a, b\}$, $\{a, c\}$, $\{a, d\}$, $\{b, c\}$, $\{b, d\}$, and $\{c, d\}$.

Combinations₂

Theorem 2: The number of r -combinations of a set with n elements, where $n \geq r \geq 0$, equals

$$C(n, r) = \frac{n!}{(n-r)!r!}.$$

Proof: By the product rule $P(n, r) = C(n, r) \cdot P(r, r)$.
Therefore

$$C(n, r) = \frac{P(n, r)}{P(r, r)} = \frac{n!/(n-r)!}{r!/(r-r)!} = \frac{n!}{(n-r)!r!}.$$

Combinations₃

Example 11: How many poker hands of five cards can be dealt from a standard deck of 52 cards? Also, how many ways are there to select 47 cards from a deck of 52 cards?

Solution: Since the order in which the cards are dealt does not matter, the number of five card hands is:

$$\begin{aligned} C(52, 5) &= \frac{52!}{5!47!} \\ &= \frac{52 \cdot 51 \cdot 50 \cdot 49 \cdot 48}{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = 26 \cdot 17 \cdot 10 \cdot 49 \cdot 12 = 2,598,960 \end{aligned}$$

The different ways to select 47 cards from 52 is

$$C(52, 47) = \frac{52!}{47!5!} = C(52, 5) = 2,598,960$$

This is a special case of a general result. →

Combinations₄

Corollary 2: Let n and r be nonnegative integers with $r \leq n$. Then $C(n, r) = C(n, n - r)$.

Proof: From Theorem 2, it follows that

$$C(n, r) = \frac{n!}{(n-r)!r!}$$

and

$$C(n, n-r) = \frac{n!}{(n-r)![n-(n-r)]!} = \frac{n!}{(n-r)!r!}.$$

Hence, $C(n, r) = C(n, n - r)$.

Combinations₅

Example: A group of 30 people have been trained as astronauts to go on the first mission to Mars. How many ways are there to select a crew of six people to go on this mission?

Solution: By Theorem 2, the number of possible crews is

$$C(30, 6) = \frac{30!}{6!24!} = \frac{30 \cdot 29 \cdot 28 \cdot 27 \cdot 26 \cdot 25}{6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = 593,775$$

Section Summary₃

Permutations

The **number** of r -permutations of a set with n elements is denoted by $P(n, r)$.

$$P(n, r) = n(n - 1)(n - 2) \cdots (n - r + 1)$$

r -permutations of a set with n distinct elements.

Combinations

The number of r -combinations of a set with n distinct elements is denoted by $C(n, r)$.

$$C(n, r) = \frac{n!}{(n - r)!r!}.$$

8. In how many different orders can five runners finish a race if no ties are allowed?

11. How many bit strings of length 10 contain

- a)** exactly four 1s?
- b)** at most four 1s?
- c)** at least four 1s?
- d)** an equal number of 0s and 1s?

$$8. \quad P(5, 5) = 5! = 120$$

11, 210 386 848 252

- 21.** How many permutations of the letters $ABCDEFGH$ contain
- a) the string BCD ?
 - b) the string $CFGA$?
 - c) the strings BA and GF ?

29. A club has 25 members.

- a) How many ways are there to choose four members of the club to serve on an executive committee?
- b) How many ways are there to choose a president, vice president, secretary, and treasurer of the club, where no person can hold more than one office?

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Homework

Page 417 25 50

Page 426 4 7

Page 435 22bc 28